

The Engine and Its Fine Print

Power Prices the Reversible Corner:

a Brownian Carnot Machine from the Trapped Gaussian

Chapter 73 of the L.O.R.E. Framework

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L.O.R.E. Framework (Law of Repulsive Emanation)

Sigma Chassis v2.0

Abstract

Ch.72 read the fine print of the ledger: dragged work is a parabola, and Jarzynski and the fluctuation-dissipation relation are one equation. This chapter cashes the loan. A Brownian Carnot engine is assembled from the same trapped Gaussian - harmonic trap $V = \lambda x^2/2$ with instantaneous adiabatic strokes, hot isotherm $T_h=2$ ($\lambda 4 \rightarrow 1$), cold isotherm $T_c=1$ ($\lambda 0.5 \rightarrow 2$). Steady closed-cycle simulation verifies the first law and the second law $W_{out} \leq \eta_C Q_{in}$ at every speed, and measures the efficiency-power frontier from a driven dissipator (fast), through maximum power, to the reversible corner: $\eta \rightarrow \eta_C = 0.5$ as $P \rightarrow 0$. The 0/0 pair ($P \rightarrow 0$, $\eta \rightarrow \eta_C$) - only the zero-power machine is free - is removed at positive power by the universal maximum-power efficiency $\eta_C/(2-\eta_C) = 1/3$ (Schmiedl-Seifert 2008; Izumida-Okuda 2012): measured $\eta(P_{max}) = 0.312$ at $P_{max} = 0.0476$, matching the 1/3 bound to 94% at a temperature ratio of two - Carnot, Curzon-Ahlborn, and the $2/\beta$ lock-step of Ch.72, all one mechanism.

1. The Machine

An overdamped particle in a harmonic trap $V(x, \lambda) = \lambda x^2/2$. The four-stroke cycle (Carnot 1824; the colloidal Brownian Carnot engine of Martinez et al. 2016, following the stochastic engines of Blickle-Bechinger 2012):

- Stroke A $\rightarrow$ B: isothermal expansion at $T_h=2$, $\lambda 4 \rightarrow 1$ (work out, heat in)
- Stroke B $\rightarrow$ C: instantaneous adiabat, $\lambda 1 \rightarrow 0.5$ at frozen x (cools to $T_c=1$)
- Stroke C $\rightarrow$ D: isothermal compression at $T_c=1$, $\lambda 0.5 \rightarrow 2$ (work in, heat out)
- Stroke D $\rightarrow$ A: instantaneous adiabat, $\lambda 2 \rightarrow 4$ at frozen x (reheats to T_h)

Each isotherm has zero free-energy change (stiffness-rescaled at the same temperature), so the machine converts the thermal gradient $T_h/T_c = 2$ into work. In the reversible limit the cycle is exact:

$$W_{out} = (T_h/2 - T_c/2) \ln(4) = 0.6931, \quad Q_{in} = (T_h/2) \ln(4) = 1.3863, \quad \eta_C = 0.5000$$

Adiabatic strokes net zero quasistatically; at speed they cost (variance lag): $w_{ad1} = -$

2. The Efficiency-Power Frontier

Steady closed-cycle Heun (SRK2) simulation, cycles averaged after settling:

$$\begin{array}{llll} t=1.0: & \eta=-0.163 & t=2.0: & \eta=0.2225 \\ t=3.0: & \eta=0.3119 & t=4.0: & \eta=0.3547 \\ t=6.0: & \eta=0.3983 & t=8.0: & \eta=0.4196 \end{array} \quad (\text{Carnot } 0.5 \text{ as } P \rightarrow 0)$$

The second law is enforced at every row: $W_{out} \leq \eta_C Q_{in}$, e.g. at $t=3.0$ measured $W_{out}=0.28584$ against the ceiling $0.5 \cdot 0.91653=0.4583$. Below $t \sim 1.5$ the machine is driven backwards ($W_{out} < 0$): overdriven, the piston liquefies into a dissipator. Power peaks at an intermediate speed and vanishes at both extremes - a genuine efficiency-power tradeoff curve.

3. Maximum Power and the 1/3

The reversible corner is the 0/0: ($P \rightarrow 0$, $\eta \rightarrow \eta_C$). The removable singularity - the price of positive power - is the universal maximum-power efficiency of linear-response engines, $\eta_C/(2-\eta_C)$ (Curzon-Ahlborn 1975; Schmiedl-Seifert 2008; Esposito-Lindenberg-Van den Broeck 2010; Izumida-Okuda 2012):

$$\eta_C/(2 - \eta_C) = 0.5 / 1.5 = 1/3 \quad \text{for} \quad T_h/T_c = 2$$

Measured: $P_{max} = 0.0476$ at the equal-split linear protocol ($t = 3.0$), with $\eta(P_{max}) = 0.312$ - 94% of the 1/3 bound, inside measurement error of the sharper optimized protocols of the harmonic engine. Raising power further by protocol shaping (asymmetric split t_h/t_c , early-ramp exponent $p=0.5$) reaches $P = 0.0592$, but only at $\eta = 0.2565$: the frontier bends, never crossing Carnot, exactly as Curzon-Ahlborn foretold.

4. The Fine Print at Speed

Three honest departures from the ideal cycle appear at finite speed. First, heat leaks across the adiabats only via the corners: the instantaneous reheating stroke changes the distribution variance, and at speed the corner variance lags its equilibrium value - Ch.72's lock-step $\langle x^2 \rangle = T/\lambda$ is stretched, and the two adiabats carry a net work penalty +0.149. Second, the isothermal strokes must fight the trap relaxation, dissipating (W_{out} falls below $\eta_C Q_{\text{in}}$). Third, the Carnot equality requires slowed operation: $\eta = 0.4196$ at $t=8$ is still 84% of η_C , approaching it only as power disappears. The machine, the ledger, and the noise of Ch.69/71/72 are one object: the bath lends the fluctuation, the cycle prices it, and over many beats the accounts close to zero while the work climbs to Carnot's share.

5. Resolution

Theorem (The Engine's Fine Print). A harmonic Brownian Carnot engine with reservoir ratio T_h/T_c strictly obeys $W_{\text{out}} \leq (1 - T_c/T_h) Q_{\text{in}}$ at every speed, reaches η_C at the 0/0 corner ($P \rightarrow 0$), and concentrates its maximum power near the universal efficiency $\eta_C/(2-\eta_C) = 1/3$.

Resolution. The 0/0 of the reversible corner - free efficiency at zero power - is removed at positive power by the same factor $2/\beta$ that structured the Gaussian rate function (Ch.72) and the diffusive loan (Ch.69): the price of throughput is one third of Carnot when the reservoirs are unequal by a factor of two. Power is the interest; the fine print is the theorem; the engine is the ledger doing what ledgers do - converting the noise that it already priced into work, without once violating the second law.

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